

backtrack.

Your GPS recalculates. Why doesn't learning?

A new route to the lesson you need now.

Try **BACKTRACK**

backtrack-learning.vercel.app

University of the Philippines Manila



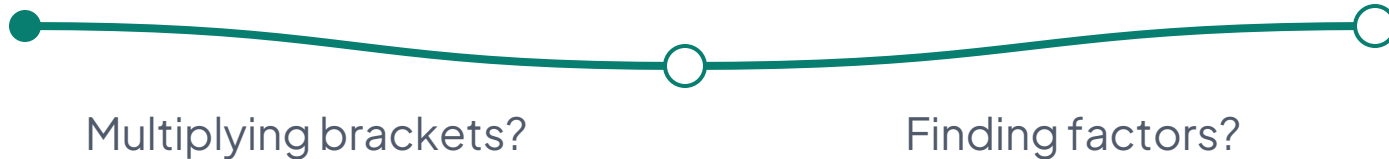
“Review.”

Review what?

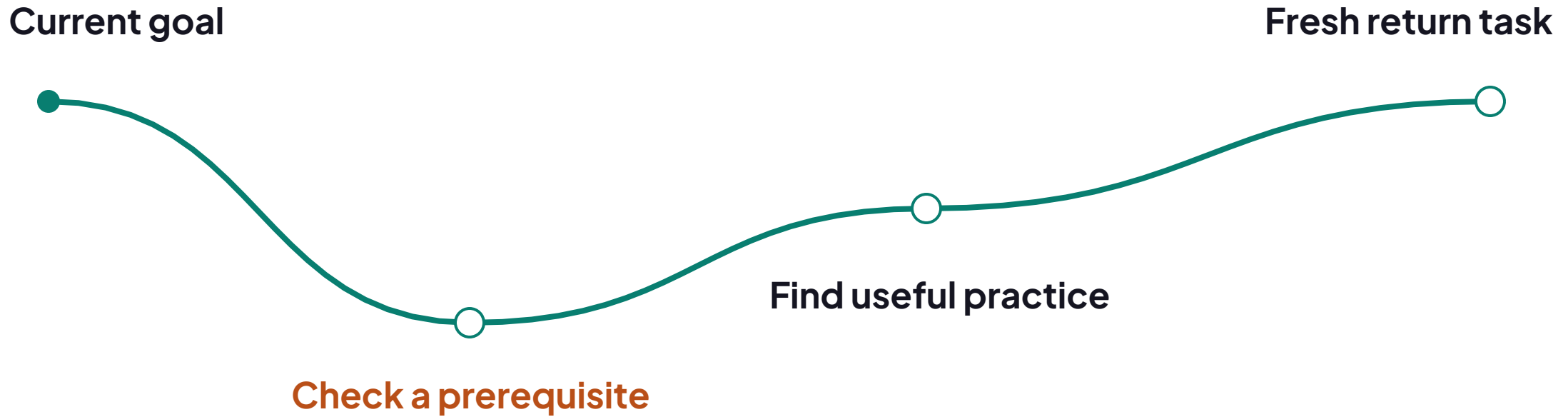
TODAY'S CLASS

$$x^2 + 7x + 12 = 0$$

A learner can be stuck here because of a step underneath.



The destination stays. The route changes.



A wrong answer opens an investigation. It is not a permanent label.

The reward is less review.

BEFORE THE CHECKS



One review removed.

Earlier progress stays.

AFTER TWO FRESH CHECKS



Try BACKTRACK

Khan is the learning engine.



01

Watch here

Official Khan lessons inside the route.

02

Practice on Khan

Matched exercises in an existing learning environment.

03

Return here

Fresh checks reconnect the route.

Back to today's lesson.

FRESH DESTINATION TASK

$$x^2 + 11x + 28 = 0$$

$$x = -4 \text{ or } x = -7$$

The route ends with something the learner can demonstrate.

Did the return last?

Retained reentry

Pass after the route.

Pass again 7–14 days later.

Report against every enrolled baseline-blocked learner.

Five months to earn the evidence.

80–120

learners across 2–3 cohorts

Planning targets, subject to agreements, permissions and access.

NOV

Review + baseline

DEC–JAN

First cycle +
comparison

FEB

Second educator

MAR

Follow-up + handover

A fair comparison.

Teacher–selected Khan practice

BACKTRACK–supported Khan practice

Comparable access time and adult support.

Fresh post and delayed tasks in both groups.

Randomized delayed start where feasible. Otherwise report feasibility, not causation.

Adaptive learning already exists.

ALEKS

Objectives and prerequisites

Gemini

Goal-based diagnosis and study plans

Khan

Personalization and learning infrastructure

Our opportunity: a repeatable Khan recovery routine that schools can inspect and test.

One goal. A manageable routine.

- 01 Teacher confirms the destination.
- 02 Learners follow different routes.
- 03 Khan assignments provide practice.
- 04 Separate supervised tasks check the return.

Measure teacher minutes, not just learner clicks.

A kit another educator can use.

01 A reviewed destination map

02 Checks and answer rationales

03 Matched Khan material

04 Parallel return assessments

05 Facilitator and reporting guide

Topic links for schools, Reels and self-study.

PHP 100,000 implementation scenario.

Learning review	24,000	Product operations	8,000
Access support	18,000	Teacher onboarding	8,000
Deployment	16,000	Recruitment content	3,000
Evaluation	18,000	Contingency	5,000

Planning allocation. Not quotes or secured funds. Learner access stays free.

Keep the goal. Change the route.

Try BACKTRACK

backtrack-learning.vercel.app

University of the Philippines Manila

Matthew Labrador · Paul Recio · Harry Gomez

Coach: Justin Mesias

A new route to today's lesson. A clear way to measure the return.

Research behind the approach

Khan teacher reports

ALEKS custom objectives

Gemini study notebooks (June 2026)

IES learning practice guide

Spacing and retrieval (2022)

Confidence assessment (Foster)

Self-Determination Theory in education